# Destynie Medeiros, PhD

Icahn School of Medicine at Mount Sinai
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## EDUCATION & TRAINING

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| 2024- Present | Icahn School of Medicine at Mount Sinai, New York, NY<br>Postdoctoral Fellow, Nash Family Department of Neurosciences<br>Advisor: Denise Cai, PhD |
| 2019-2024 | University of Alabama at Birmingham, Birmingham, AL<br>PhD in Neuroscience, Department of Neuroscience<br>Advisor: Lucas Pozzo-Miller, PhD |
| 2015-2019 | University of Hartford, Hartford, CT<br>Bachelor of Arts, Biology, Summa Cum Laude with Honors Thesis<br>Advisor: Paola Sacchetti, PhD |

## GRANTS & FELLOWSHPS

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|---|---|
| 2026-2029 | Junior Fellow, Simons Foundation ($524,645)<br>*Circuit mechanisms of memory integration* |
| 2026 | NIH F32 MH144887 ($232,428)<br>(*declined for Simons Fellowship)*<br>*Circuit mechanisms of memory integration* |
| 2023-2024 | NIH T32 NS061788 ($58,728)<br>*Insights into Rett syndrome: Social Behavior, BDNF-TrkB Signaling, and Dendritic Spines* |
| 2022-2023 | NIH Diversity Supplement R01MH118563-04S1 ($64,449)<br>*Role of the Hippocampal-mPFC Pathway in Social Memory Deficits in Autism* |

## PUBLICATIONS

*co-first authors

Acevedo-Triana C, **Medeiros D**, Lopes Goncalez J, Li W, Pozzo-Miller L. Atypical social behaviors in mouse models for Rett syndrome (2026). *Frontiers in Neurology*. PMID: 42064979.

Li H, Tomar R, Chen TJ, Scinasi A, Zorab JM, **Medeiros D**, Cai DJ, Schaffer ES, Wu X. Top-down control of neuromodulation orchestrates social allostasis (2026). Under review at Nature Neuroscience.

**Medeiros D,** Polepalli L, Li W, Pozzo-Miller L. Altered activity of mPFC pyramidal neurons and parvalbumin-expressing interneurons during social interactions in a *Mecp2* mouse model for Rett syndrome (2025). *Journal of Neurophysiology*, PMID: 39149275.

**Medeiros D**, Ayala-Baylon K, Edigo-Betancourt H, Miller E, Chapleau C, Robinson H, Phillips, ML, Yang T, Longo, FM, Li W, Pozzo-Miller L. A small-molecule TrkB ligand improves dendritic spine phenotypes and atypical behaviors in female Rett syndrome mice (2024). *Disease Models & Mechanisms*, PMID: 38785269.

***Featured interview in Disease Models and Mechanisms: First person- Destynie Medeiros***

***Disease Models and Mechanisms Outstanding Paper Prize 2024***

**Medeiros D***, McMurry K*, Pfeiffer M, Newson K, Testerman T, Graf J, Silver AC, Sacchetti P. Slowing Alzheimer's disease progression through probiotic supplementation (2024). *Frontiers in Neuroscience*. PMID: 38510467.

Li C*, Saliba NB*, Martin H, Losurdo NA, Kolahdouzan K, Siddiqui, **Medeiros D**, Li W; "Purkinje cell dopaminergic inputs to astrocytes regulate cerebellar-dependent behavior (2023). *Nature Communications*, PMID: 36959176.

Sakamoto S*, Mallah D*, **Medeiros DJ**, Dohi E, Imai T, Rose IVL, Matoba K, Zhu X, Kamiya A, Kano SI. Alterations in circulating extracellular vesicles underlie social stress-induced behaviors in mice" (2021). *FEBS Open Bio*, PMID: 34043886.

***Wiley Top Cited Article 2021-2022***

Abele SH*, Meadows KE*, **Medeiros D***, Silver A. Time is on the Immune System's Side, Yes it is." (2019). *Yale Journal of Biology and Medicine.* PMID: 3124983.

## AWARDS & HONORS

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|---|---|
| 2026-2029 | Junior Fellow, Simons Foundation |
| 2026-2028 | Associate, Neuroscience Scholars Program, Society for Neuroscience |
| 2026 | NIH F32 Postdoctoral Fellowship (*declined for Simons Junior Fellow)* |
| 2026 | Anchor Award for Outstanding Alumni: Early Career Professional, University of Hartford |
| 2026 | Carl Storm Fellowship, Gordon Research Conference, Optogenetic Approaches to Understanding Neural Circuits and Behavior |
| 2026 | Invited Attendee, Flatiron Institution Neural Data Analysis Workshop, Simons Foundation |
| 2024 | Outstanding Paper Prize, *Disease Models and Mechanisms* |
| 2024 | Finalist, Wu Tsai Postdoctoral Fellowship, Yale University *(declined)* |
| 2023-2024 | NIH T32 Predoctoral Fellowship |
| 2023 | Invited Speaker, Rutgers Brain Health Institute Postdoc Recruitment Event, Rutgers University (*declined due to schedule conflict)* |

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| 2022-2023 | Travel Award, Civitan International Research Center, University of Alabama at Birmingham |
| 2022-2023 | NIH R01 Diversity Supplement |
| 2022 | Travel Award, Graduate Student Government Association, University of Alabama at Birmingham |
| 2021-2022 | Wiley Top Cited Article Prize |
| 2019 | Belle K. Ribicoff Prize for Exemplary Graduating Senior, University of Hartford |
| 2018 | Elisabeth Swain Biology Award for Academic Achievement, University of Hartford |
| 2018 | Travel Award, Faculty for Undergraduate Neuroscience (FUN) |
| 2017-2019 | Dorothy Goodwin Women's Advancement Initiative Scholarship, University of Hartford |
| 2017-2019 | College of Arts and Sciences Research Grant, University of Hartford |
| 2017 | Student Government Association Research Grant, University of Hartford |
| 2014 | The Veterans of Foreign Wars (VFW) Scholarship (*merit-based*), Fall River, MA |

**INIVTED TALKS**

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| 2026 | Optogenetic Approaches to Understanding Neural Circuits and Behavior Gordon Research Seminar, Lucca Italy |
| 2025 | Simons Foundation, NY |
| 2024 | Icahn School of Medicine at Mount Sinai |
| 2024 | New York University |
| 2023 | Cognition and Cognitive Disorders T32 Data Blitz, University of Alabama at Birmingham |
| 2019 | National Conference on Undergraduate Neuroscience (NCUR), Atlanta Georgia |
| 2019 | Honors College Undergraduate Colloquium, University of Hartford |
| 2018 | Honors College Undergraduate Colloquium, University of Hartford |

**INSTITUTIONAL TALKS**

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| 2026 | Center for Neurotechnology and Behavior, Icahn School of Medicine at Mount Sinai |
| 2024 | Neurobiology Department Works-In-Progress, University of Alabama at Birmingham |
| 2020-2023 | Neurobiology Summer Seminar Series, University of Alabama at Birmingham |

**POSTER PRESENTATIONS** **#equal contribution*

1. **Medeiros D***, van der Rijn E*, Baggetta AM, Durand-de Cottoli R, Russo SJ, Cai DJ#, Pennington ZT#. Chronic social defeat induces stress sensitization and anterior hypothalamic nucleus response. Optogenetic Approaches to Understanding Neural Circuits and Behaviors, Gordon Research Conference, Lucca, Italy. May 2026.

2. **Medeiros D***, van der Rijn E*, Baggetta AM, Durand-de Cottoli R, Russo SJ, Cai DJ#, Pennington ZT#. Chronic social defeat induces stress sensitization and anterior hypothalamic nucleus response. Optogenetic Approaches to Understanding Neural Circuits and Behaviors, Gordon Research Seminar, Lucca, Italy. May 2026.
3. **Medeiros D**, Zaki Y, Ko B, Bacon ME, Cai DJ. Aversive memory integration across the dorsoventral hippocampal axis through brain states. Friedman Brain Institute Annual Retreat, New York, NY. May 2025.
4. **Medeiros D**, Polepalli L, Li W, Pozzo-Miller L. Hippocampal-prefrontal neuronal dynamics contributing to social memory deficits in a mouse model for Rett syndrome. International Rett syndrome Foundation Research Meeting, Westminster, CO. June 2024.
5. **Medeiros D**, Pozzo-Miller L. Hippocampal-prefrontal neuronal dynamics contributing to social memory deficits in a mouse model for Rett syndrome. McKnight Brain Research Foundation Poster Reception, Washington, DC. November 2023.
6. **Medeiros D**, Pozzo-Miller L. Hippocampal-prefrontal neuronal dynamics contributing to social memory deficits in a mouse model for Rett syndrome. Society for Neuroscience Annual Meeting, Washington, DC. November 2023.
7. **Medeiros D**, Pozzo-Miller L. Hippocampal-prefrontal neuronal dynamics contributing to social memory deficits in a mouse model for Rett syndrome. Molecular & Cellular Cognition Society Annual Meeting, Washington, DC. November 2023.
8. **Medeiros D**, Ayala-Baylon K, Edigo-Betancourt H, Miller E, Chapleau C, Robinson H, Phillips, ML, Yang T, Longo, FM, Li W, Pozzo-Miller L. A small-molecule TrkB ligand improves dendritic spine phenotypes and atypical behaviors in female Rett syndrome mice. National Enhancement of Underrepresented Academic Leaders (NEURAL) Conference, Birmingham, AL. June 2023.
9. **Medeiros D**, Ayala-Baylon K, Edigo-Betancourt H, Miller E, Chapleau C, Robinson H, Phillips, ML, Yang T, Longo, FM, Li W, Pozzo-Miller L. A small-molecule TrkB ligand improves dendritic spine phenotypes and atypical behaviors in female Rett syndrome mice. International Rett syndrome Foundation Research Meeting, Nashville, TN. June 2023.
10. **Medeiros D**, Ayala-Baylon K, Edigo-Betancourt H, Miller E, Chapleau C, Robinson H, Phillips, ML, Yang T, Longo, FM, Li W, Pozzo-Miller L. A small-molecule TrkB ligand improves dendritic spine phenotypes and atypical behaviors in female Rett syndrome mice. Society for Neuroscience Annual Meeting, San Diego, CA. November 2022.
11. **Medeiros D**, Ayala-Baylon K, Edigo-Betancourt H, Miller E, Chapleau C, Robinson H, Phillips, ML, Yang T, Longo, FM, Li W, Pozzo-Miller L. A small-molecule TrkB ligand improves dendritic spine phenotypes and atypical behaviors in female Rett syndrome mice. Molecular & Cellular Cognition Society Annual Meeting, San Diego, CA. November 2022.
12. **Medeiros D**, Ayala-Baylon K, Edigo-Betancourt H, Miller E, Chapleau C, Robinson H, Phillips, ML, Yang T, Longo, FM, Li W, Pozzo-Miller L. A small-molecule TrkB ligand improves dendritic spine phenotypes and atypical behaviors in female Rett syndrome mice. University of Alabama at Birmingham Comprehensive Neuroscience Center Annual Retreat, Guntersville, AL. September 2022.
13. **Medeiros D,** Buitrago N, Lyons C, Silver A, Sacchetti P. Effects of Probiotics on Alzheimer's Disease Pathology. Women's Advancement Initiative's Annual Symposium, University of Hartford. April 2019.

## TEACHING

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| 2025 | Guest Lecturer, *In vivo Calcium Imaging,* Icahn School of Medicine at Mount Sinai (ISMMS) |
| 2025 | Teaching Assistant, Advanced Neurotechniques Course, Max Planck Florida Institute |
| 2018, 2019 | Instructor, Genetics Laboratory, University of Hartford |
| 2017-2019 | Biology and Chemistry Tutor, University of Hartford |

## MENTORSHIP

### ROLES

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|---|---|
| 2024-present | Graduate Student, Associate Researcher Mentor, ISMMS |
| 2025 | High School Student Mentor, ISMMS |
| 2022-2023 | Near Peer Mentor, University of Alabama at Birmingham (UAB) |
| 2020-2021 | Undergraduate Student Mentor, UAB |
| 2021 | Blazer BRAIN Student Mentor, UAB |

### DIRECT MENTEES

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| 2024-present | Elijah van der Rijn, Graduate Student, ISMMS |
| 2024-present | Ann Pierre-Louis, Associate Researcher, ISMMS |
| 2026 | Afra Mahmud, Undergraduate Research Assistant, Hunter College |
| 2025 | Charles Lipschultz, High School Volunteer, ISMMS |
| 2024-2025 | Madeline Bacon, Associate Researcher, ISMMS<br>Current: PhD Neuroscience Student, University of California, Los Angeles |
| 2023 | Nehal Pagadala, Undergraduate Neuroscience Summer Program, UAB |
| 2022-2023 | Likhitha Polepalli, Associate Researcher, UAB<br>Current: PhD Medical Physics Student, University of Wisconsin, Madison |
| 2021 | Drew Bookman, Blazer BRAIN Program for Underrepresented Trainees in Neuroscience, UAB<br>Current: Medical Student, Louisianna State University Health Science Center |
| 2020 | Jana Badrani, Rotation Student, UAB<br>Current: MD/PhD Student, UAB |

## LEADERSHIP & SERVICE

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| 2025 | Conference Organizer, Inaugural Conference, New York Memory Hub |
| 2024-present | Junior Executive Committee Chair, New York Memory Hub |
| 2024 | Junior Executive Committee Founder, New York Memory Hub |
| 2021-2024 | Committee Member, Neurobiology Seminar Series Planning Committee University of Alabama at Birmingham |
| 2021-2024 | Recruitment Volunteer, Graduate Biomedical Sciences Program, University of Alabama at Birmingham |
| 2017-2019 | Undergraduate Chair, Brain Awareness Week Committee, University of Hartford, Greater Hartford Coalition |
| 2017-2019 | Events Coordinator, Pre-Professional Club, University of Hartford |